

SAFETY DATA SHEET (GHS)

Propylene (Propene) | CAS: 115-07-1 | Formula: C₃H₆

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Propene, CAS 115-07-1 (<https://echa.europa.eu/>)
- PubChem CID 8252 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/8252>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0681 — Propylene
- NOAA CAMEO Chemicals — Propylene datasheet (<https://cameochemicals.noaa.gov/>)
- Supplier SDS — Propylene (industrial/refrigerated grade), multiple suppliers
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Propylene (Propene)

CAS Number: 115-07-1

Chemical formula: C₃H₆

Molecular weight: 42.08 g/mol

Synonyms: Propene; Methylethylene; 1-Propene

Intended use: Primary feedstock — alkylation unit feed to produce cumene (reaction with benzene). PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Gas 1 (H220); Press. Gas — Liquefied gas (H280)

Signal word: Danger

Hazard statements (H-codes): H220 (Extremely flammable gas); H280 (Contains gas under pressure; may explode if heated)

Precautionary statements (P-codes): P210, P211, P251, P261, P271, P377, P381, P403, P405, P410+P403

GHS pictograms: GHS02 (Flammable), GHS04 (Gas cylinder)

Other hazards: Handled as a refrigerated/pressurised liquefied gas — major inventory of flammable gas under pressure represents a significant fire/explosion (BLEVE/jet fire/vapour cloud) hazard for the alkylation feed system. Heavier than air as a vapour — can accumulate in low-lying areas, drains and enclosed spaces, displacing oxygen (simple asphyxiant at high concentrations) and travelling to remote ignition sources. Rapid release of liquefied propylene causes cold burns/frostbite from auto-refrigeration.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Propylene: CAS 115-07-1 — chemical/polymer grade typically >99% purity; confirm purity specification and trace propane/ethylene content against PPCL alkylation feed specification (trace propane affects relief/flare loading calculations)

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air immediately. High concentrations displace oxygen — if victim is unconscious, remove using rescue equipment/SCBA (do not enter without protection). Administer oxygen if breathing is difficult; perform artificial respiration if breathing has stopped. Obtain medical attention.

Skin contact: In case of contact with liquefied gas, areas of frostbite/cold burn should be flushed with lukewarm water (do not rub); remove non-adhered clothing. Obtain medical attention for cold burns.

Eye contact: In case of liquid/cold gas contact causing frostbite, flush gently with lukewarm water for at least 15 minutes. Obtain medical attention.

Ingestion: Not applicable — material is a gas/liquefied gas under normal handling; ingestion exposure route is not relevant

Notes to physician: Treat as simple asphyxiant exposure (assess for hypoxia) and/or cryogenic/cold-contact injury (frostbite) as applicable. No specific antidote.

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, water spray/fog for cooling surrounding equipment and dispersing vapours; do not extinguish a leaking gas fire unless the leak can be stopped — extinguishing the flame without stopping the leak creates an unconfined vapour cloud explosion hazard

Unsuitable extinguishing media: None specifically — but see special hazards regarding extinguishing a flowing gas fire

Special hazards: Extremely flammable gas (H₂20) with very wide flammable range (LEL 2.0%, UEL 11.1%).

Pressurised/liquefied storage — vessels exposed to fire may undergo BLEVE (boiling liquid expanding vapour explosion). Vapours are heavier than air and travel to ignition sources, with possible flash back. Auto-refrigeration on release can cause metal embrittlement (low-temperature) issues for non-cryogenic-rated components.

Special protective equipment for firefighters: Full bunker gear with positive-pressure SCBA; cool fire-exposed vessels/pipework with water spray from maximum distance using unmanned monitors where possible; evacuate area if fire cannot be controlled and a BLEVE risk exists

Hazardous combustion products: Carbon monoxide, carbon dioxide

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Evacuate area immediately; eliminate all ignition sources over a wide radius (vapour travel distance can be significant). Approach release from upwind/uphill. Use SCBA for any approach within the vapour cloud. Be aware of low-oxygen atmospheres in confined/low-lying areas.

Environmental precautions: Primarily an air-quality/fire-explosion concern rather than aquatic — propylene is a gas at ambient conditions and disperses rapidly if not ignited

Containment and cleanup: Stop the source of release if it is safe to do so (remote isolation valves). Allow gas to disperse/vent to a safe location; use water spray to disperse vapour clouds and protect personnel/structures, with caution as this can also spread the flammable cloud. Do not attempt to extinguish a sustained jet fire from a leaking source unless flow can be stopped.

Reference to other sections: See Section 7 (Handling/Storage) for pressurised system controls and Section 9 (flammable range data)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Handle only in closed systems designed for pressurised/liquefied flammable gas service. Use only explosion-proof/intrinsically safe electrical equipment in classified areas. Bond and ground transfer equipment. Avoid mechanical/thermal shock to pressurised equipment.

Storage conditions: Store in pressure vessels/bullets rated for the design pressure and temperature, with appropriate relief devices (PSVs) discharging to a safe location (flare). Protect storage vessels from fire exposure (fireproof insulation/water deluge per facility fire protection design) to mitigate BLEVE risk.

Temperature limits: Maintain within design temperature/pressure envelope of storage vessels; propylene boils at -47.6°C at atmospheric pressure — refrigerated storage (if used) requires materials rated for cryogenic service

Pressure: Stored/handled as a liquefied gas under its own vapour pressure (critical temperature -91.8°C, critical pressure -4.6 MPa) — relief system design must account for fire case and blocked-outlet scenarios

Segregation requirements: Segregate from oxidisers, sources of ignition, and incompatible materials. Maintain separation distances from occupied buildings/ignition sources per facility hazardous area classification.

Special requirements: Continuous flammable gas detection in the alkylation feed area; remotely operable emergency isolation valves (EIVs); pressure relief valves vented to flare; fireproofing/water deluge on storage vessels per fire risk assessment

Specific end-use recommendations: As the primary alkylation feedstock, ensure metering/control valve trains have adequate leak detection and that the reactor feed system is designed for the relief/blowdown scenarios associated with a large flammable gas inventory

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No established ACGIH TLV or OSHA PEL for propylene specifically (treated as a simple asphyxiant — ACGIH recommends maintaining oxygen levels above 18% by volume rather than a substance-specific limit). AIHA Workplace Environmental Exposure Level (WEEL) of 500 ppm (TWA) has historically been referenced by some sources — confirm current applicability.

TWA: Not formally established (simple asphyxiant — manage via oxygen monitoring and ventilation rather than a substance TWA)

STEL: Not established

Ceiling: Not established

Skin notation: Not applicable

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: Closed system handling; continuous combustible gas (LEL) detection in alkylation unit and storage areas; adequate ventilation in enclosed/below-grade spaces; oxygen monitoring for confined space entry

PPE — Respiratory: Not required for normal operations in open/ventilated areas; SCBA required for any entry into an area where a release has occurred or oxygen deficiency is possible

PPE — Gloves: Insulated/cryogenic-rated gloves for any task with potential contact with liquefied propylene (cold burn protection)

PPE — Eyes: Safety glasses for routine tasks; face shield/goggles for tasks with potential for liquid release (cold burn/frostbite protection)

PPE — Body: Flame-resistant (FR) clothing; insulated/cryogenic PPE for liquid-contact tasks

Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless gas (liquefied under pressure for transport/storage)

Odour: Faint petroleum-like/sweet odour (odourised commercial product is not typical for chemical-grade feedstock — confirm odourisation status for area monitoring strategy)

Odour threshold: ~20 ppm (varies by source) — odour is not a reliable warning property at flammable concentrations

pH: Not applicable

Melting point / Freezing point: -185.2°C

Boiling point (at 1 atm): -47.6°C

Flash point: -108°C (extremely low — gas at ambient conditions)

Evaporation rate: Not applicable (gas)

Flammability (solid/gas): Extremely flammable gas

Flammability limits in air: LEL 2.0 vol%; UEL 11.1 vol%

Vapour pressure (at 20°C): ~880 kPa (~8.7 atm) — significant pressure when liquefied at ambient temperature

Vapour density (air = 1): ~1.5 (heavier than air)

Relative density (water = 1): ~0.51 (liquid, at boiling point — much lighter than water)

Solubility (water): ~200 mg/L at 25°C (slightly soluble)

Partition coefficient (log Kow): ~1.77

Auto-ignition temperature: ~455°C

Decomposition temperature: Not applicable under normal conditions

Viscosity: Not applicable (gas); liquid viscosity very low (~0.1 mPa·s at boiling point)

Explosive properties: Not classified as explosive substance; forms explosive vapour/air mixtures within LEL-UEL range

Oxidising properties: Not an oxidiser

Molecular weight: 42.08 g/mol

Critical temperature / pressure: ~91.8°C / ~4.6 MPa — relevant to relief system design for storage vessels

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; reactive as an alkene — can polymerise or react with strong oxidisers, halogens, and certain catalysts (this reactivity is the basis of the intended alkylation process)

Chemical stability: Stable under recommended storage and handling conditions; uncontrolled polymerisation is not typically a concern at ambient conditions without catalyst

Possibility of hazardous reactions: Can react vigorously with strong oxidisers (fire/explosion risk); reacts with halogens (e.g. chlorine, bromine) sometimes violently; forms explosive mixtures with air across a wide concentration range

Conditions to avoid: Heat, sparks, open flames, and other ignition sources; mechanical/thermal shock to pressurised equipment; contact with strong oxidisers and halogens outside of controlled process conditions

Incompatible materials: Strong oxidisers, halogens, strong acids (can catalyse polymerisation/side reactions)

Hazardous decomposition products: None under normal storage; combustion produces CO and CO₂

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation (primary — asphyxiation/anaesthetic effects at high concentration); dermal/eye contact with liquefied gas (cold burns)

Acute toxicity — Oral LD₅₀: Not applicable (gas)

Acute toxicity — Dermal LD₅₀: Not applicable (gas); cold contact causes physical injury (frostbite), not chemical toxicity

Acute toxicity — Inhalation LC₅₀: Low acute toxicity — primary acute hazard is asphyxiation (oxygen displacement) and mild CNS/anaesthetic effects (dizziness, drowsiness) at very high concentrations; not classified for acute inhalation toxicity under GHS

Skin corrosion/irritation: Not classified — physical cold-contact injury (frostbite) is the relevant hazard from liquefied gas, not chemical irritation

Serious eye damage/irritation: Not classified for chemical eye damage; cold-contact injury from liquefied gas is the relevant hazard

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified as a carcinogen

Reproductive toxicity: Not classified

STOT (single exposure): High concentrations can cause dizziness, drowsiness and asphyxiation due to oxygen displacement; not formally classified under STOT criteria

STOT (repeated exposure): Not classified

Aspiration hazard: Not applicable (gas)

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC₅₀/EC₅₀): Low aquatic toxicity expected — propylene is a gas at ambient conditions with low water solubility and rapidly volatilises from water; not classified for aquatic hazards under GHS

Persistence and degradability: Rapidly degraded in the atmosphere via reaction with OH radicals (short atmospheric half-life); not persistent

Bioaccumulative potential: Low (volatile gas, low log K_{ow})

Mobility in soil: Highly mobile as a gas — volatilises rapidly to atmosphere

Other adverse effects: Minor contributor to photochemical smog formation (reactive alkene, high VOC reactivity) — relevant to air emissions permitting

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via controlled venting to flare/thermal oxidiser per facility procedures; do not vent directly to atmosphere in quantities that create a flammable atmosphere or exceed permit limits

Contaminated packaging: Return cylinders/vessels to supplier per program; ensure depressurised and purged before any maintenance

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535; site air emissions permit (flare/VOC provisions)

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1077

Proper shipping name: PROPYLENE

Transport hazard class(es): Class 2.1 (Flammable gas)

Packing group: Not applicable (gases)

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: Transport only in vessels rated for the substance's vapour pressure; ensure pressure relief devices are functional and vented to a safe location; bonding/grounding during transfer

Bulk transport (MARPOL/IBC): Classify per IGC Code if shipped in bulk by sea as liquefied gas

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 115-07-1)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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